

Learning Polyhedral Conformal Sets for Robust Optimization

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Abstract

Robust optimization (RO) provides a principled framework for decision-making under uncertainty, but its performance critically depends on the choice of the uncertainty set. While large sets ensure reliability, they often lead to overly conservative decisions, whereas small sets risk excluding the true outcome. Recent data-driven approaches, particularly conformal prediction, offer finite-sample validity guarantees but remain largely task-agnostic, ignoring the downstream decision structure. In this paper, we propose a decision-aware conformal framework that learns uncertainty sets tailored to robust optimization objectives. Our approach parameterizes a flexible family of polyhedral sets via data-driven hyperplanes and learns their geometry by directly minimizing the induced robust loss, while preserving statistical validity through conformal calibration. To correct for data-dependent selection, we incorporate a re-calibration step on an independent dataset to restore coverage. The resulting sets capture directional and anisotropic uncertainty aligned with the decision objective while remaining computationally tractable. We provide finite-sample coverage guarantees and bounds on the sub-optimality gap to an oracle decision. This work bridges the gap between statistical validity and decision optimality, providing a principled framework for data-driven robust optimization.

1 Introduction

Robust optimization (RO) has emerged as a fundamental framework for decision-making under uncertainty [Ben-Tal and Nemirovski, 1998, Ben-Tal et al., 2009], with widespread applications across operations research, finance, energy systems, supply chain management, and machine learning. At its core, RO prescribes decisions that perform well under worst-case realizations of uncertain parameters, offering a principled way to hedge against ambiguity and model misspecification. This paradigm is particularly valuable in high-stakes settings, where

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poor decisions can lead to significant economic loss or societal impact. For example, power grid operators must plan infrastructure upgrades under uncertain demand and renewable generation [Abdin et al., 2022, Jabr, 2013, Zhou and Zhu, 2024]; supply chain managers must allocate inventory without knowing future disruptions [Bertsimas and Thiele, 2006]; and healthcare systems must make treatment decisions under uncertain patient outcomes [Mo et al., 2021]. In such settings, robustness is not merely desirable—it is essential.

A central component of robust optimization is the specification of the uncertainty set, which characterizes plausible realizations of the unknown parameters. The quality of the resulting decision critically depends on this choice. A well-known challenge is the trade-off between *conservativeness and performance*: overly large uncertainty sets provide strong protection against worst-case scenarios but often lead to overly pessimistic decisions, while overly small sets may exclude the true outcome, resulting in unreliable and potentially catastrophic decisions.

While the literature has extensively studied how to ensure robustness—often by constructing uncertainty sets that are sufficiently large to guarantee feasibility or coverage—it has paid comparatively less attention to the risk of over-conservativeness [Wang et al., 2025a, Zhou and Zhu, 2025]. In practice, this imbalance can have significant consequences. For instance, in power systems planning, an overly conservative uncertainty set may trigger unnecessary infrastructure investments, leading to excessive costs for utilities and consumers [Chen et al., 2026b]. In inventory management, it may result in systematic overstocking, tying up capital and increasing waste [Lim and Wang, 2017]. Similarly, in portfolio optimization, overly conservative risk sets can lead to excessively defensive allocations that sacrifice substantial returns [Gregory et al., 2011]. These examples highlight the importance of striking the right balance: uncertainty sets should be large enough to ensure reliability, yet sufficiently tight and structured to avoid overly pessimistic decisions.

Recent advances have sought to address the construction of uncertainty sets through data-driven approaches, among which conformal prediction has gained particular attention [Yeh et al., 2024]. Conformal methods provide distribution-free, finite-sample guarantees for prediction sets under mild assumptions [Vovk et al., 2005], making them an appealing tool for robust optimization. By using conformal prediction sets as uncertainty sets, one can ensure that the true outcome is contained within the set with high probability, thereby offering a principled way to control risk. However, standard conformal prediction is inherently *task-agnostic*: it constructs sets based solely on predictive accuracy, without accounting for the downstream decision problem [Wang et al., 2023]. As a result, conformal uncertainty sets may be unnecessarily large or poorly shaped from the perspective of decision-making, leading to suboptimal robust decisions. Several recent works have attempted to improve the efficiency of conformal sets by reducing their size or adapting them to the data distribution [Zheng and Zhu, 2024]. While these methods yield tighter prediction sets, they remain largely disconnected from the structure of the downstream optimization problem. Consequently, there remains a fundamental gap between *statistical validity* and *decision optimality*: existing approaches either guarantee coverage without considering decision performance, or improve efficiency without explicitly optimizing for the decision task.

In this paper, we bridge this gap by proposing a decision-aware conformal prediction framework for robust optimization. The central idea is to design uncertainty sets whose geometry is explicitly aligned with the structure of the downstream decision problem, rather

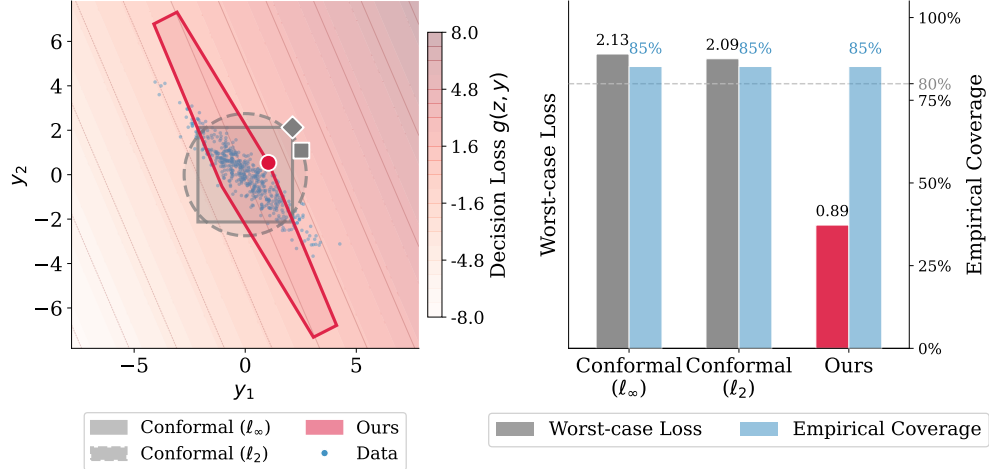


Figure 1: Illustration of decision-aware conformal prediction sets. Standard conformal sets (grey) and our decision-aware set (red), overlaid with the loss $g(z, y) = z^\top y$. Both sets achieve the same valid coverage (85%), but our set is oriented to avoid high-loss directions, reducing worst-case decision loss.

than treating them as task-agnostic byproducts of prediction (Fig. 1). To this end, we introduce a flexible, learnable nonconformity score that induces *polyhedral uncertainty sets* via data-driven hyperplanes. This construction enables the uncertainty set to capture *directional and anisotropic uncertainty*, prioritizing directions that are most relevant to the decision objective. At the same time, the polyhedral structure preserves computational tractability, allowing the resulting robust optimization problem to admit efficient reformulations. Building on this formulation, our method jointly learns the geometry of the uncertainty set by minimizing the induced robust loss, while maintaining statistical validity through conformal calibration. Since the uncertainty set is learned from data, we incorporate a re-calibration step on an independent dataset to correct for selection bias and restore finite-sample coverage guarantees. Our framework provides both statistical and decision-theoretic guarantees. We establish finite-sample coverage for the final, data-dependent uncertainty set, addressing the post-selection bias. We further derive finite-sample bounds on the sub-optimality gap relative to an oracle decision, with a decomposition into calibration and learning errors. This result quantifies the effect of finite samples and provides guidance on allocating data between learning and re-calibration.

Related Work Adapting conformal prediction (CP) methods to construct valid uncertainty sets in robust optimization (RO) has been a key focus of recent literature development in CP. Some works propose using different shapes, such as using ellipsoids [Johnstone and Cox, 2021] and multiple generated balls [Patel et al., 2024b], to accurately carve the shape of the data and prevent overconservatism. Other works find applications of CP in various real-world RO problems, such as power system resilience [Chen et al., 2025, 2026b] and autonomous driving [Lekeufack et al., 2024]. Another line of literature studies the properties of CP methods in different types of RO problems, such as linear-quadratic regulator systems [Patel et al., 2024a], human decision making [Hullman et al., 2025], inverse optimiza-

tion problems [Lin et al., 2024], and decision-theoretic perspectives [Chenreddy and Delage, 2024, Kiyani et al., 2025] to derive optimality conditions for CP methods under RO settings and gain implementational insights. Our work is closely related to this line of research, as we also aim to tailor conformal prediction to better serve robust optimization. However, we differ in our approach: rather than deriving theoretical characterizations of the uncertainty set, we adopt a learning-based framework to estimate its optimal shape.

Tailoring conformal prediction (CP) sets to be more efficient (*i.e.*, tighter) is another important direction of CP research. Previous works have proved optimal to construct sets in high-density regions (HDR) of the underlying target distribution [Izbicki et al., 2022, Lei and Wasserman, 2014], which motivates a series of work studying the use of generative models as prediction models to generate multiple fine-grained sets whose union approximates the contour of the HDR Wang et al. [2023], Zheng and Zhu [2024]. On the other hand, multidimensional CP focuses on developing more expressive nonconformity scores that allow the prediction set to take flexible shapes such as ellipsoid [Xu et al., 2024], convex [Tumu et al., 2024], or even nonconvex shapes [Fang et al., 2025]. More recently, several directions of work have explored how conformal prediction set shapes can be explicitly optimized for. This can be formulated as a model selection problem [Liang et al., 2024, Yang and Kuchibhotla, 2025], where the task is to select the best performing model from a finite collection of conformal prediction sets. Another line of work formulates this as an end-to-end learning task whose objective is to minimize set size subject to the validity chance-constraint, and can be solved using, *e.g.*, empirical risk minimization [Bai et al., 2022, Braun et al., 2025] and duality [Kiyani et al., 2024]. Our work is most closely related to this line of research, as we also formulate a learning problem. However, rather than optimizing an efficiency-oriented objective as in prior works, we consider a general decision-making objective that subsumes efficiency as a special case.

Our work is closely related to decision-focused learning (DFL) [Mandi et al., 2024, Wilder et al., 2019], a class of approaches that train predictive models by optimizing decision-oriented objectives through differentiable optimization layers [Agrawal et al., 2019, Amos and Kolter, 2017], and has been adapted to settings such as generative modeling [Wang et al., 2025b] and sequential modeling [Chen et al., 2026a]. The idea of DFL can be extended to learn conformal prediction models tailored for robust optimization, in a way similar to efficiency-aware conformal prediction [Stutz et al., 2022]. However, this field remains under-explored. For example, Cortes-Gomez et al. [2024] proposes decision-aware nonconformity scores that incorporate downstream utility into conformal prediction, yielding sets aligned with decision objectives while preserving coverage guarantees. Yeh et al. [2024] considers a more general framework by parameterizing convex uncertainty sets via neural scores and training them end-to-end using differentiable optimization layers. Bao et al. [2025] instead studies a model selection approach, selecting from a discrete set of conformal predictors to minimize downstream decision risk. Our work shares the same motivation as tailoring conformal prediction to downstream decision-making while maintaining validity. However, rather than designing specific scores, relying on end-to-end training, or selecting from a finite model class, we adopt a learning-based approach that directly optimizes the shape of the polyhedron conformal prediction set for a general decision objective to achieve optimality.

2 Problem Setup and Preliminaries

In many high-stakes decision-making problems, the true outcome $y \in \mathcal{Y} \subseteq \mathbb{R}^d$ is unknown at the time a decision $z \in \mathcal{Z} \subseteq \mathbb{R}^p$ must be made, where $\mathcal{Z}$ denotes the feasible decision region. A standard approach is to first construct an uncertainty (or prediction) set $\mathcal{C}(x) \subseteq \mathcal{Y}$ based on observed features $x \in \mathcal{X}$, and then select a decision that is robust against all realizations within this set [Johnstone and Cox, 2021]. Formally, given a loss function $g(z, u)$ that evaluates the performance of decision z under a potential outcome $u \in \mathcal{Y}$ in context x , the decision is obtained by solving the contextual robust optimization problem

$$\min_{z \in \mathcal{Z}} \max_{u \in \mathcal{C}(x)} g(z, u),$$

where the inner maximization captures adversarial uncertainty over the set $\mathcal{C}(x)$, and the outer minimization selects the decision that minimizes the worst-case loss.

A key challenge in this framework lies in constructing the uncertainty set $\mathcal{C}(x)$. In particular, there is an inherent *trade-off between robustness and informativeness*: overly large sets guarantee coverage but often lead to overly conservative decisions with high loss, whereas overly small sets may fail to contain the true outcome, resulting in unreliable or even infeasible decisions [Zhou and Zhu, 2025].

To address this challenge, we seek to construct uncertainty sets that are both *statistically valid* and *decision-aware*. Specifically, we consider a parameterized family of set-valued mappings $\{\mathcal{C}(\cdot; \theta) : \mathcal{X} \rightarrow 2^{\mathcal{Y}}\}$ indexed by $\theta \in \Theta$. Let $(X, Y) \sim \mathcal{P}$ denote the joint distribution of features and outcomes. Our goal is to learn a parameter θ such that: (i) the resulting uncertainty set achieves a desired coverage level, *i.e.*, $\mathbb{P}\{Y \in \mathcal{C}(X; \theta)\} \geq 1 - \alpha$, and (ii) the induced robust decision incurs low worst-case loss.

This leads to the following constrained learning problem:

$$\min_{\theta \in \Theta} \ell(\theta) := \min_{z \in \mathcal{Z}} \max_{u \in \mathcal{C}(x; \theta)} g(z, u) \tag{1}$$

$$\text{s.t. } \mathbb{P}\{Y \in \mathcal{C}(X; \theta)\} \geq 1 - \alpha. \tag{2}$$

For a fixed context x and parameter θ , the inner min-max problem is a deterministic robust optimization problem that defines the decision induced by the uncertainty set $\mathcal{C}(x; \theta)$. The outer optimization over θ then selects, among all sets satisfying the coverage constraint (2), the one that yields the best downstream decision performance. Randomness enters only through the data-generating distribution $\mathcal{P}$ used to construct and evaluate $\mathcal{C}(\cdot; \theta)$, and the decision problem is fully deterministic. Thus, (1) can be viewed as a decision-focused formulation that chooses the geometry of the uncertainty set to directly optimize decision quality.

2.1 Preliminaries: Conformal Prediction

Conformal prediction provides a general, model-agnostic framework for constructing prediction sets with finite-sample, distribution-free coverage guarantees [Angelopoulos and Bates, 2023, Shafer and Vovk, 2008, Vovk et al., 2005]. Suppose the data $\{(X_i, Y_i)\}_{i=1}^{n+1}$ are *exchangeable*. Given calibration data $\mathcal{D}_n := \{(X_i, Y_i)\}_{i=1}^n$, the goal is to construct a set-valued predictor $\mathcal{C}(x) \subseteq \mathcal{Y}$ such that, for a new test point (X_{n+1}, Y_{n+1}) , the coverage guarantee

$$\mathbb{P}\{Y_{n+1} \in \mathcal{C}(X_{n+1}; \mathcal{D}_n)\} \geq 1 - \alpha \tag{3}$$

holds for a prescribed miscoverage level $\alpha \in (0, 1)$.

Assumption 1 (Exchangeability). *The augmented sample $\{(X_i, Y_i)\}_{i=1}^{n+1}$ is exchangeable, i.e., for any permutation π of $\{1, \dots, n+1\}$,*

$$(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1}) \stackrel{d}{=} (X_{\pi(1)}, Y_{\pi(1)}), \dots, (X_{\pi(n+1)}, Y_{\pi(n+1)}).$$

A common implementation is split conformal prediction [Lei et al., 2018, Vovk et al., 2005]. The data are partitioned into a training set and a calibration set. A predictive model $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ is first trained on the training set, and a nonconformity score function $s(x, y)$ is defined to measure how well a candidate outcome y conforms to the prediction at $\hat{f}(x)$. For example, in regression, one may take $s(x, y) = |y - \hat{f}(x)|$. Given the calibration data $\mathcal{D}_n$, the empirical quantile of the nonconformity scores is defined as

$$R(\mathcal{D}_n) = \inf \left\{ r : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i) \leq r\} \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\}. \quad (4)$$

The prediction set for a new input x is then constructed as

$$\mathcal{C}(x; \mathcal{D}_n) = \{u \in \mathcal{Y} : s(x, u) \leq R(\mathcal{D}_n)\}.$$

By construction, the set $\mathcal{C}$ satisfies the marginal coverage guarantee in (3) under the exchangeability assumption, regardless of the choice of the underlying predictive model $\hat{f}$.

3 Decision-Aware Conformal Calibration

We propose a decision-aware conformal framework for constructing uncertainty sets tailored to downstream optimization tasks. See Fig. 2 for an overview. Our approach parameterizes a flexible family of polyhedral sets via data-driven hyperplanes and learns their geometry by directly minimizing the induced robust decision loss. Statistical validity is ensured through conformal calibration, which provides finite-sample coverage guarantees for any fixed parameter, and is restored after model selection via a post-hoc re-calibration step on an independent dataset.

3.1 Learnable Polyhedral Conformal Set

Unlike classical uncertainty sets with fixed and isotropic geometry (e.g., ℓ_2 balls) [Ben-Tal and Nemirovski, 1998, 1999, Shafer and Vovk, 2008], our core idea is to construct a parameterized family of polyhedral sets defined as intersections of data-driven half-spaces (supporting hyperplanes), allowing the geometry to adapt to the downstream decision objective (Fig. 2a). We require statistical validity to hold uniformly over all hyperplane configurations, and then optimize the parameters within this valid family to minimize downstream decision loss.

Let $\epsilon(x, y) := y - \hat{f}(x)$ denote the residual, and define its augmented form $\tilde{\epsilon}(x, y) := [\epsilon(x, y)^\top, 1]^\top \in \mathbb{R}^{d+1}$. We parameterize the nonconformity score as

$$s(x, y; \theta) := \max_{k \in [K]} \{\theta_k^\top \tilde{\epsilon}(x, y)\}, \quad (5)$$

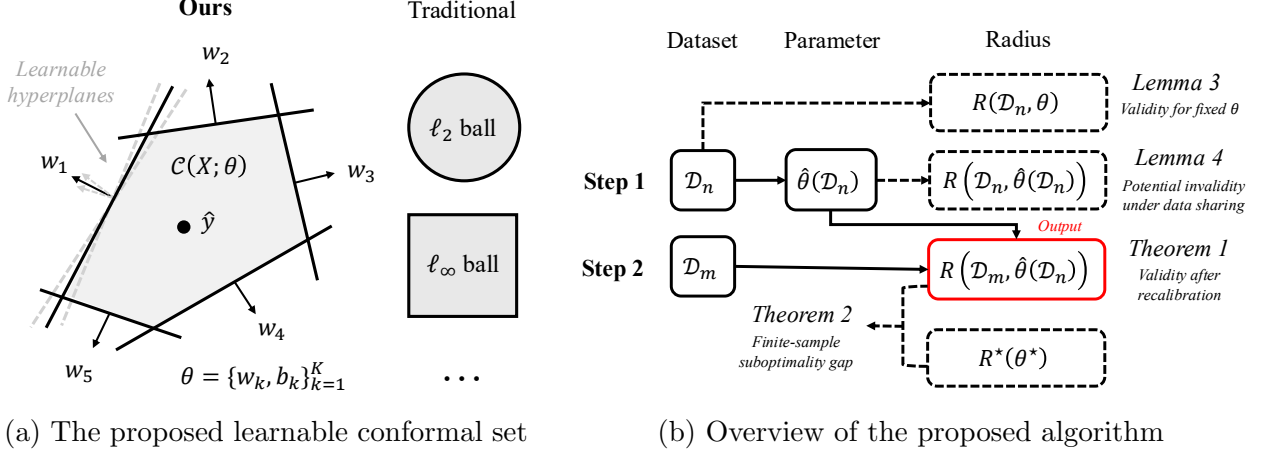


Figure 2: Overview of decision-aware conformal set learning.

where $\theta = \{\theta_k := [w_k, b_k]^\top \in \mathbb{R}^{d+1}\}_{k=1}^K$. This score is a max-affine (polyhedral convex) function, which can be interpreted as a polyhedral gauge on the residual space, whose sublevel sets define polyhedral regions shaped by the hyperplanes $\{\theta_k\}_{k=1}^K$; see, *e.g.*, [Niculescu and Persson, 2006]. With symmetric choices of $\{\theta_k\}$, it recovers classical polyhedral norms, for example, $\{\pm e_j\}_{j=1}^d$ yields the ℓ_∞ norm.

Given $s(x, y; \theta)$, we define a parameter-dependent radius via split conformal calibration on the calibration dataset $\mathcal{D}_n$:

$$R(\mathcal{D}_n; \theta) = \inf \left\{ r \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i; \theta) \leq r\} \geq \frac{[(n+1)(1-\alpha)]}{n} \right\}. \quad (6)$$

For any fixed θ and a new input x , the resulting conformal prediction set

$$\mathcal{C}(x; \mathcal{D}_n, \theta) := \{u \in \mathcal{Y} : s(x, u; \theta) \leq R(\mathcal{D}_n; \theta)\}. \quad (7)$$

satisfies the coverage guarantee in (3). To ensure that $\mathcal{C}(\cdot; \mathcal{D}_n, \theta)$ is a bounded polyhedron (polytope) for all x , we require $K \geq d+1$ and that the collection of normal vectors $\{w_k\}_{k=1}^K \subset \mathbb{R}^d$ —corresponding to the first d components of θ_k —positively spans $\mathbb{R}^d$.

3.2 Proposed Algorithm

We now present a two-stage procedure that integrates decision-aware learning of the proposed conformal set with a post-hoc re-calibration step that ensures valid coverage (Fig. 2b).

Step 1: Decision-Aware Conformal Set Learning Given the parameterized conformal set $\mathcal{C}(\cdot; \mathcal{D}_n, \theta)$ in (7), we learn the parameter θ by directly minimizing the downstream decision loss induced by the corresponding robust optimization problem:

$$\hat{\theta}(\mathcal{D}_n) := \arg \min_{\theta \in \Theta} \min_{z \in \mathcal{Z}} \max_{u \in \mathcal{Y}} g(z, u) \quad (8)$$

$$\text{s.t. } s(x, u; \theta) \leq R(\mathcal{D}_n; \theta). \quad (9)$$

Here, we explicitly denote the dependence of $\hat{\theta}$ on the calibration data $\mathcal{D}_n$.

Compared to the original formulation (1)–(2), the coverage constraint is enforced implicitly through conformal calibration via $R(\mathcal{D}_n; \theta)$. This allows the probabilistic constraint to be replaced by the deterministic constraint (9), thereby decoupling statistical validity from the optimization procedure. Moreover, the polyhedral structure induced by $s(\cdot, \cdot; \theta)$ implies that the feasible set in (9) can be represented as a finite collection of linear inequalities.

Step 2: Re-calibration for Robust Decision-Making The conformal calibration in (6) guarantees valid coverage for any *fixed* parameter θ . However, in Step 1, $\hat{\theta}$ is learned from the data $\mathcal{D}_n$, which introduces a data-dependent selection effect [Hegazy et al., 2025, Liang et al., 2024] and violates the exchangeability condition required for conformal validity.

To restore validity, we perform a re-calibration step using an independent dataset. Specifically, let $\mathcal{D}_m := \{(X_j, Y_j)\}_{j=1}^m$ denote an additional set of exchangeable samples drawn from the same distribution. Given the learned parameter $\hat{\theta}(\mathcal{D}_n)$, we compute the conformal radius on this fresh calibration set, yielding $R(\mathcal{D}_m; \hat{\theta})$. The resulting conformal set $\mathcal{C}(x; \mathcal{D}_m, \hat{\theta})$ is then used to prescribe the final decision

$$\hat{z}(x) = \arg \min_{z \in \mathcal{Z}} \max_u g(z, u), \quad \text{s.t. } u \in \mathcal{C}(x; \mathcal{D}_m, \hat{\theta}(\mathcal{D}_n)). \quad (10)$$

This re-calibration step is essential to ensure valid post-selection coverage. Conformal guarantees rely on exchangeability conditional on a fixed nonconformity score; however, optimizing θ using data breaks this condition. Intuitively, reusing the same data for both tuning θ and calibrating the radius leads to overfitting of the score function and overly optimistic (*i.e.*, too small) uncertainty sets, resulting in undercoverage. By recalibrating on an independent dataset, we restore the required independence and recover valid coverage guarantees for the final decision.

3.3 Computational Methods

To operationalize the proposed algorithm, we derive tractable reformulations of the decision-aware conformal set learning problem in (8)–(9) for linear and more general convex objective functions g . Two main computational challenges arise: (i) the nonconformity score in (5) introduces a nonsmooth max-affine constraint, and (ii) the conformal radius $R(\mathcal{D}_n; \theta)$ is an empirical quantile that depends on θ through a nonsmooth order statistic. All proofs are included in Appendix A.

We begin by reformulating the max-affine score constraint in (9) as a linear system for fixed conformal radius r . Recall that $\theta_k = [w_k^\top, b_k]^\top$, where $w_k \in \mathbb{R}^d$ and $b_k \in \mathbb{R}$, define

$$W := [w_1, \dots, w_K]^\top \in \mathbb{R}^{K \times d}, \quad b := [b_1, \dots, b_K]^\top \in \mathbb{R}^K, \quad c_n := \lceil (n+1)(1-\alpha) \rceil.$$

Lemma 1 (Linear reformulation of the score constraint). *For any radius r , the constraint $s(x, y; W, b) \leq r$ is equivalent to the following system of linear inequalities*

$$W\epsilon(x, y) + b \leq r\mathbf{1}_K,$$

where $\mathbf{1}_K$ denotes the all-one vector in $\mathbb{R}^K$.

To capture the dependence of the conformal radius $R(\mathcal{D}_n; W, b)$ on (W, b) using big- M method [Nemhauser and Wolsey, 1988, Williams, 2013], we introduce an auxiliary scalar $r \in \mathbb{R}$ and binary variables $t = [t_1, \dots, t_n]^\top \in \{0, 1\}^n$.

Lemma 2 (Big- M representation of the conformal radius). *Suppose $r \geq \underline{r}$ and $M > 0$ is chosen such that, for all feasible (W, b) and all $i \in [n]$, $W\epsilon_i + b \leq (\underline{r} + M)1_K$. Then the empirical conformal radius can be represented as*

$$\begin{aligned} R(\mathcal{D}_n; W, b) = \min_{r, t} \quad & r \\ \text{s.t.} \quad & W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \\ & 1_n^\top t \geq c_n, \quad t \in \{0, 1\}^n. \end{aligned} \tag{11}$$

Here, $t_i = 1$ enforces that the i th nonconformity score is no larger than r , whereas $t_i = 0$ relaxes the corresponding constraint. The constraint $1_n^\top t \geq c_n$ therefore requires at least c_n nonconformity scores to be bounded by r , and minimizing r recovers the empirical conformal quantile.

Combining Lemma 1 and 2 gives the following exact reformulations to problem (8)–(9).

Proposition 1 (Reformulation of problem (8)–(9)). *The problem (8)–(9) is equivalent to*

$$\begin{aligned} \min_{W, b, z, r, t} \quad & \max_{u \in \mathcal{Y}} g(z, u) \\ \text{s.t.} \quad & W \left(u - \hat{f}(x) \right) + b \leq r1_K, \\ & W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \\ & 1_n^\top t \geq c_n, \quad z \in \mathcal{Z}, \quad t \in \{0, 1\}^n. \end{aligned} \tag{12}$$

Corollary 1 (Reformulation for linear objective). *Suppose $g(z, y) = g_0(z) + z^\top Qy$. The constrained min-max problem in (12) admits the following equivalent reformulation.*

$$\begin{aligned} \min_{W, b, z, \lambda, r, t} \quad & g_0(z) + \left(r1_K - b + W\hat{f}(x) \right)^\top \lambda \\ \text{s.t.} \quad & W^\top \lambda = Q^\top z, \quad \lambda \in \mathbb{R}_+^K, \\ & W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \\ & 1_n^\top t \geq c_n, \quad z \in \mathcal{Z}, \quad t \in \{0, 1\}^n. \end{aligned} \tag{13}$$

The mixed-integer formulation in (13) can be solved exactly using spatial branch-and-bound solvers such as Gurobi [Gurobi Optimization, LLC, 2026] for small K and n . For general concave objectives $g(z, \cdot)$ in (12), we can employ Column-and-Constraint Generation (CCG) [Zhao and Zeng, 2012], which alternates between a master problem over (W, b, z, r, t) and an adversarial subproblem over u . See Alg. 1 in Appendix A.5 for details. However, the master problem remains a mixed-integer program with exponential complexity on the calibration size n [Nemhauser and Wolsey, 1988]. When $g(z, \cdot)$ is convex, the adversarial subproblem over the polyhedral conformal set attains an optimum at extreme point, but enumerating all vertices can be computationally demanding in high dimensions: a polytope in $\mathbb{R}^d$ defined by K halfspaces can have $\mathcal{O}(K^{\lfloor d/2 \rfloor})$ vertices.

To obtain a more efficient solution, we relax the hard empirical quantile constraint (9) using a continuous penalty with pinball loss [Steinwart and Christmann, 2011]. This formulation is computationally preferable for two reasons: (i) It replaces the n binary variables in (11) with a differentiable penalty. (ii) It enables stochastic gradient methods. The gradient of each iteration can be estimated with subgradients using minibatches of calibration samples, significantly reducing the number of constraints per iteration and thus complexity. See Alg. 2 and Appendix A.5 for the full implementation.

4 Theoretical Analysis

This section presents two central theoretical properties: (i) the re-calibrated parameter yields a valid conformal prediction set, and (ii) the suboptimality gap admits a finite-sample characterization under mild regularity conditions. The key assumptions and proofs are deferred to Appendix B.

4.1 Statistical Validity of the Re-calibrated Conformal Set

We begin by recalling the standard validity guarantee of conformal prediction for a fixed nonconformity score, which is well established in the literature [Angelopoulos and Bates, 2023].

Lemma 3 (Validity for Fixed θ). *Under the exchangeability assumption, for any fixed $\theta \in \Theta$, the conformal prediction set $\mathcal{C}_\theta(\cdot; \alpha)$ defined in (7)–(6) satisfies*

$$\mathbb{P}\{Y_{n+1} \in \mathcal{C}(X_{n+1}; \mathcal{D}_n, \theta)\} \geq 1 - \alpha.$$

However, in our algorithm, θ is estimated from the calibration data, yielding a data-dependent choice $\hat{\theta}(\mathcal{D}_n)$. This breaks the exchangeability underlying conformal validity: for a fixed score, the augmented scores are exchangeable and the test rank is uniform, but when the score is fitted to the calibration data, this symmetry no longer holds. As a result, the prediction set tends to overfit the calibration samples, often becoming too small and leading to undercoverage.

The following result formalizes this breakdown. Its proof follows arguments similar to Theorem 2 by Barber et al. [2023] and Corollary 1 by Zhou et al. [2025]. Let $d_{\text{TV}}(\mathcal{D}_{n+1}, \mathcal{D}_{n+1}^{(i)} | \mathcal{D}_n)$ denote the total variation distance between the conditional distributions of the augmented sample $\mathcal{D}_{n+1} = \{(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})\}$ and the swapped sequence $\mathcal{D}_{n+1}^{(i)}$, obtained by exchanging the $(n+1)$ -th and i -th entries, given $\mathcal{D}_n$. Intuitively, this quantity measures the degree to which exchangeability is violated after conditioning on the calibration data. Under perfect exchangeability, swapping the test point with any calibration point leaves the conditional distribution unchanged, yielding zero total variation distance. Therefore, larger values indicate stronger asymmetry induced by data reuse, corresponding to dependence between the learned parameter $\hat{\theta}(\mathcal{D}_n)$ and the nonconformity scores, which may lead to undercoverage.

Lemma 4 (Potential Invalidity Under Data Sharing). *Under the exchangeability assumption, for any fixed $\theta \in \Theta$, if the learning data are reused in constructing the conformal prediction set $\mathcal{C}(\cdot, \mathcal{D}_n; \theta)$, then:*

$$\mathbb{P}\{Y_{n+1} \in \mathcal{C}(X_{n+1}; \mathcal{D}_n, \hat{\theta}(\mathcal{D}_n))\} \geq 1 - \alpha - \mathbb{E} \left[d_{\text{TV}} \left(\mathcal{D}_{n+1}, \mathcal{D}_{n+1}^{(i)} \mid \mathcal{D}_n \right) \right].$$

Our algorithm prevents such invalidity via a re-calibration step using an independent split dataset, thereby restoring finite-sample validity even when θ is data-dependent.

Theorem 1 (Validity of the Re-calibrated Set). *Suppose the re-calibration data $\{(X_j, Y_j)\}_{j=1}^m$, the test point (X_{m+1}, Y_{m+1}) , and the data used to obtain $\hat{\theta}$ are mutually exchangeable. Then the re-calibrated prediction set satisfies*

$$\mathbb{P}\{Y_{m+1} \in \mathcal{C}(X_{m+1}; \mathcal{D}_m, \hat{\theta}(\mathcal{D}_m))\} \geq 1 - \alpha.$$

4.2 Sub-optimality Gap of the Prescribed Decision

We now analyze the optimality of the proposed method by quantifying the gap between the decision induced by the learned parameter $\hat{\theta}$ and that of an oracle choice θ^* . The goal is to characterize how estimation, calibration, and optimization errors propagate to the final decision, and to establish finite-sample bounds on the resulting suboptimality gap.

To formalize this analysis, we first introduce oracle benchmarks and a radius-explicit formulation of the robust loss. For a fixed x and any θ , define the oracle conformal radius

$$R^*(\theta) = \inf_r \{r \in \mathbb{R} : \mathbb{P}\{s(X, Y; \theta) \leq r\} \geq 1 - \alpha\},$$

and the corresponding oracle parameter

$$\theta^* = \arg \min_{\theta} \min_{z \in \mathcal{Z}} \max_u g(z, u), \quad \text{s.t. } s(x, u; \theta) \leq R^*(\theta). \quad (14)$$

To make the dependence on the conformal radius explicit, we define the robust loss

$$\mathcal{L}(\theta, r) = \min_{z \in \mathcal{Z}} \max_{u: s(x, u; \theta) \leq r} g(z, u). \quad (15)$$

Theorem 2 (Finite-Sample Suboptimality Gap). *Let $\hat{\theta}$ be the estimated parameter obtained by solving (8)–(9), and let θ^* be the oracle parameter by solving (14). Let $\mathcal{L}(\cdot)$ be the robust loss defined in (15), and denote the suboptimality gap as: $\Delta := \mathcal{L}(\hat{\theta}(\mathcal{D}_n), R(\mathcal{D}_m; \hat{\theta}(\mathcal{D}_n))) - \mathcal{L}(\theta^*, R^*(\theta^*))$. Then, under Assumption 2, there is with probability at least $1 - \delta$:*

$$\Delta \leq \frac{L_r}{\kappa} \sqrt{\frac{\log(4/\delta)}{2m}} + \frac{2L_r}{\kappa} \sqrt{\frac{32[V \log(en/V) + \log(16/\delta)]}{n}} + \epsilon_{\text{opt}}.$$

In $\mathcal{O}_{\mathbb{P}}$ notation [Van der Vaart, 2000], the bound simplifies to

$$\Delta \leq \mathcal{O}_{\mathbb{P}} \left(\sqrt{\frac{1}{m}} + \sqrt{\frac{\log n}{n}} \right).$$

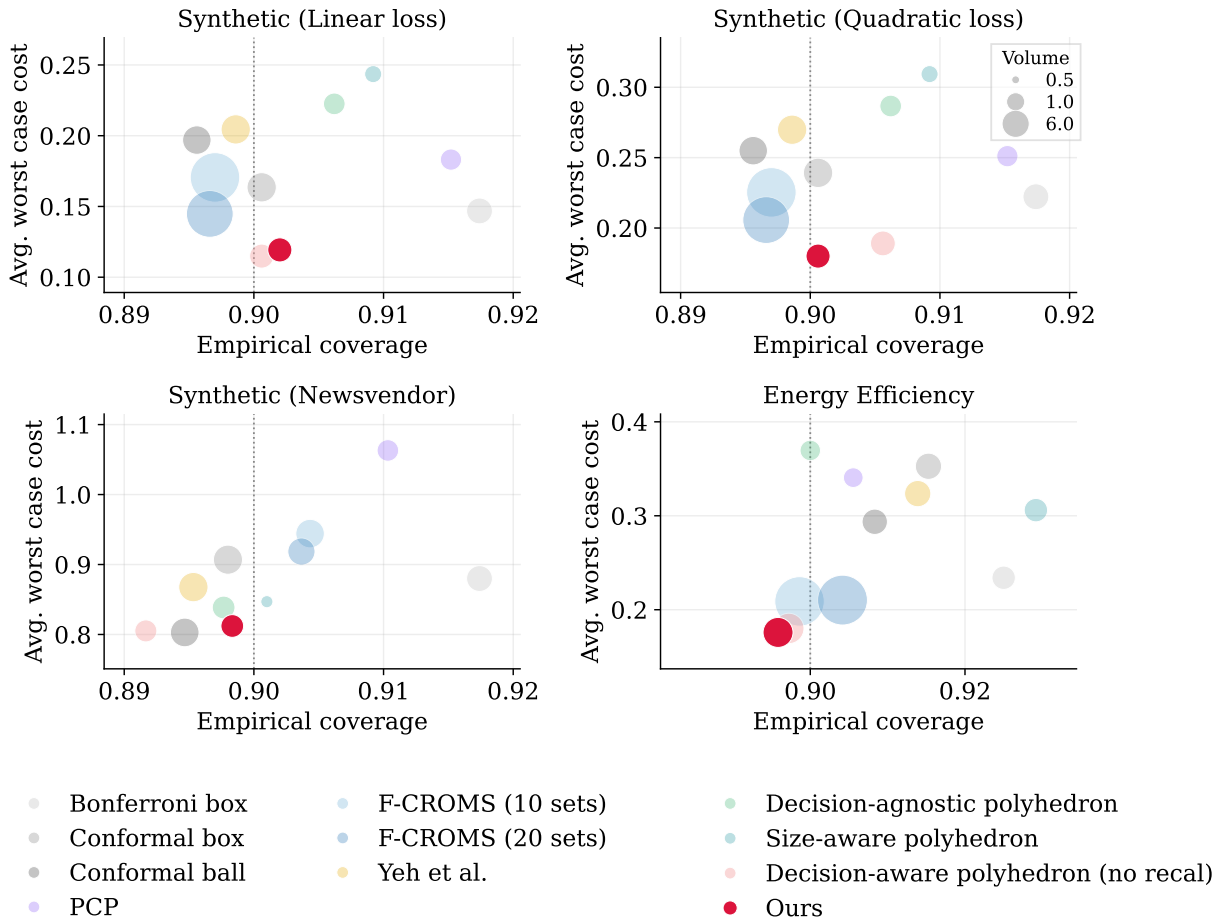


Figure 3: The proposed method in comparison with baselines. The horizontal axis shows the empirical coverage rate; the vertical axis presents the average worst-case cost; and bubble size for the volume of uncertainty.

5 Experiments

We evaluate the proposed decision-aware conformal framework on robust decision problems using both synthetic and real-data tasks. Across all experiments, we first fit a predictor $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ on an independent training split and construct residuals $\epsilon_i = Y_i - \hat{f}(X_i)$. The remaining data are split into a learning set $\mathcal{D}_n$, used to learn the set geometry $\hat{\theta}(\mathcal{D}_n)$, and an independent calibration set $\mathcal{D}_m$, used only for the final conformal radius. This yields the final set $\mathcal{C}(x; \mathcal{D}_m, \hat{\theta}(\mathcal{D}_n))$, from which robust decisions are computed on held-out test covariates. Baselines that do not learn set geometry use $\mathcal{D}_n \cup \mathcal{D}_m$ as their calibration set.

We compare against decision-agnostic conformal sets, including Bonferroni boxes [Dunn, 1961], conformal boxes or balls [Vovk et al., 2005], and PCP [Wang et al., 2023], as well as decision-aware alternatives based on F-CROMS [Bao et al., 2025] and PICNN scores [Yeh et al., 2024]. We also include three ablations of our method: a polyhedral set without decision-aware learning, a size-aware polyhedron that uses volume as the loss function, and our decision-aware polyhedron without the post-selection re-calibration step. We evaluate

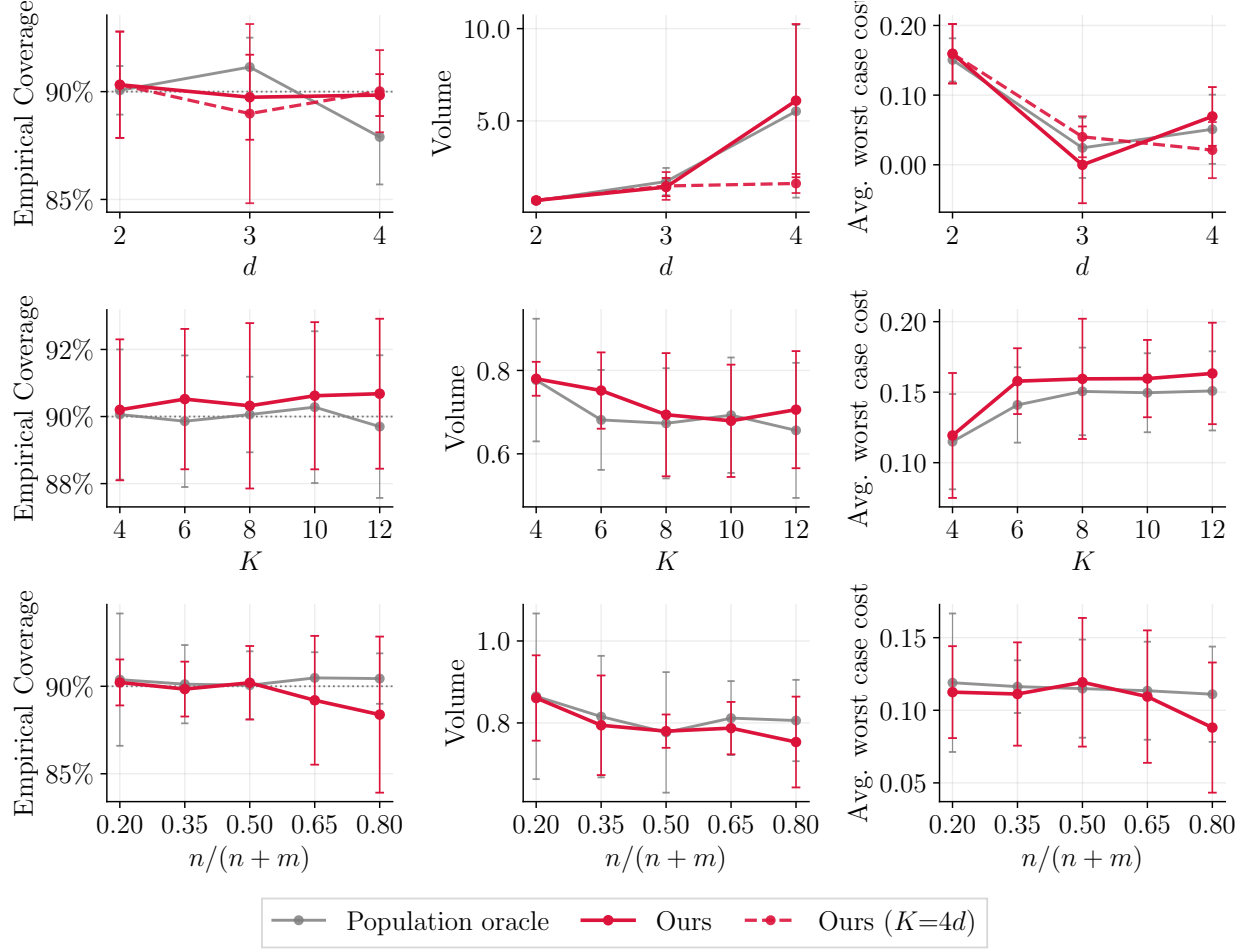


Figure 4: Hyperparameter sensitivity of coverage, volume, and worst-case cost. Ours uses $K = 8$ in the first row and $K = 4$ in the last row.

empirical coverage, average worst-case cost, average regret, set volume, and wall-clock time. Additional details on the objective function, datasets, and implementation are deferred to Appendix C.1. The experiments reported in this section are averaged over five random runs.

Main results. Fig. 3 and Tables 1 and 2 summarize the results. Across all tasks, our re-calibrated decision-aware polyhedron attains empirical coverage close to the nominal 90% level, confirming that the independent calibration split restores validity after learning the set geometry. Among valid methods, our method consistently achieves competitive or lowest worst-case cost. The comparison with PCP [Wang et al., 2023] and the size-aware polyhedron shows that sharper or smaller uncertainty sets are not necessarily better for downstream robust optimization: although both often produce smaller-volume sets, this does not consistently translate into lower worst-case cost or regret, as minimizing volume can remove directions that are important for lowering the worst-case decision loss. Compared with F-CROMS [Bao et al., 2025], adding more candidate sets of hyperplanes can reduce regret, but its model-selection approach remains less flexible than ours because it selects among a

finite candidate class rather than directly learning the conformal-set geometry. The ablation without re-calibration highlights the importance of post-selection calibration. Although its worst-case cost is often close to that of the re-calibrated method, its coverage can fall below the target level because the same data are used both to select the set geometry and to calibrate the radius, consistent with Lemma 4. By contrast, the proposed two-stage procedure restores coverage with little loss in downstream performance.

Sensitivity analysis. Fig. 4 studies the sensitivity of our method to d , K , and $n/(n+m)$. As d increases, coverage remains close to the nominal 90% level, while volume grows to satisfy the joint coverage. Scaling the number of hyperplanes with $K = 4d$ helps control the volume growth and stabilizes worst-case cost, suggesting that the polyhedral template should be more expressive in higher dimensions. Varying K has only a mild effect on coverage. Larger K slightly reduces volume at start, while worst-case cost empirical coverage remains largely stable. Finally, varying $n/(n+m)$ shows expected learning–calibration tradeoff: allocating more data to $\mathcal{D}_n$ improves geometry learning and can reduce cost, consistent with Theorem 2. But leaving too few samples in $\mathcal{D}_m$ also makes conformal radius less stable and can lead to coverage below the nominal level.

6 Conclusion and Discussion

We introduced a decision-aware conformal framework for robust optimization that learns uncertainty sets by optimizing downstream robust loss while preserving finite-sample coverage through independent re-calibration. The polyhedral parameterization captures directional uncertainty while admitting tractable robust reformulations. We established validity for the final learned set and derived a suboptimality-gap bound that separates calibration and learning errors, showing that conformal sets can be designed for both coverage and decision quality. A limitation is the focus on polyhedral geometry, which may be less suitable for highly nonlinear uncertainty. Future work could extend this framework to richer neural or flow-based set families while maintaining calibration validity and computational tractability.

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A Proofs in Section 3.3

A.1 Proof of Lemma 1

Proof. By definition of the max-affine nonconformity score,

$$s(x, y; \theta) = \max_{k \in [K]} \{w_k^\top \epsilon(x, y) + b_k\}.$$

Therefore, for any $r \in \mathbb{R}$,

$$s(x, y; \theta) \leq r \iff w_k^\top \epsilon(x, y) + b_k \leq r, \quad \forall k \in [K].$$

Stacking these K inequalities gives

$$W\epsilon(x, y) + b \leq r1_K.$$

□

A.2 Proof of Lemma 2

Proof. By definition in Eq. 6,

$$\begin{aligned} R(\mathcal{D}_n; \theta) &= \inf \left\{ r \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{s(X_i, Y_i; \theta) \leq r\} \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\} \\ &= \min \{ r \in \mathbb{R} : \sum_{i=1}^n \mathbb{1}(s(X_i, Y_i; \theta) \leq r) \geq c_n \}. \end{aligned} \tag{16}$$

By Lemma 1, $s(X_i, Y_i; \theta) \leq r$ can be expressed as

$$W\epsilon_i + b \leq r1_K = (r + M(1 - t_i))1_K$$

with large M when $t_i = 1$. Therefore,

$$\begin{aligned} R(\mathcal{D}_n; W, b) &= \min_{r, t} r \\ \text{s.t. } & W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \\ & 1_n^\top t \geq c_n, \quad t \in \{0, 1\}^n. \end{aligned} \tag{17}$$

□

A.3 Proof of Proposition 1

Define auxiliary variable r . The worst-case value

$$\max_{u \in \mathcal{Y}} g(z, u) \tag{18}$$

$$\text{s.t. } s(x, u; \theta) \leq r. \tag{19}$$

is nondecreasing in r , because increasing r enlarges the feasible region of the inner maximization. Hence, problem (8)–(9) is equivalent to:

$$\min_{\substack{\theta, z, r \\ r \geq R(\mathcal{D}_n; \theta)}} \max_{\substack{u \in \mathcal{Y} \\ s(x, u; \theta) \leq r}} g(z, u). \quad (20)$$

By Lemma 1, the score constraint in the inner maximization layer admits the linear representation

$$s(x, u; W, b) \leq r \iff W(u - \hat{f}(x)) + b \leq r1_K.$$

It remains to represent the $r \geq R(\mathcal{D}_n; W, b)$ in outer minimization layer. By Lemma 2, for a valid choice of M :

$$R(\mathcal{D}_n; W, b) = \min_{r, t} \left\{ r : W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \quad 1_n^\top t \geq c_n, \quad t \in \{0, 1\}^n \right\}.$$

Equivalently, $r \geq R(\mathcal{D}_n; W, b)$ holds if and only if there exists $t \in \{0, 1\}^n$ such that

$$W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \quad 1_n^\top t \geq c_n.$$

Therefore, (20) is equivalent to:

$$\begin{aligned} & \min_{W, b, z, r, t} \max_{u \in \mathcal{Y}} g(z, u) \\ & \text{s.t.} \quad W(u - \hat{f}(x)) + b \leq r1_K, \\ & \quad W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \\ & \quad 1_n^\top t \geq c_n, \quad z \in \mathcal{Z}, \quad t \in \{0, 1\}^n. \end{aligned} \quad (21)$$

This is exactly (12).

A.4 Proof of Corollary 1

Proof. By Proposition 1, the robust part of (12) can be written as

$$\min_{W, b, z, r, t} \left\{ \max_u \left\{ g_0(z) + z^\top Qy : W(u - \hat{f}(x)) + b \leq r1_K \right\} \right\},$$

together with the calibration constraints

$$W\epsilon_i + b \leq (r + M(1 - t_i))1_K, \quad i \in [n], \quad 1_n^\top t \geq c_n, \quad t \in \{0, 1\}^n.$$

Since $g_0(z)$ does not depend on y , the inner maximization is equivalent to

$$g_0(z) + \max_u \left\{ z^\top Qu : W(u - \hat{f}(x)) + b \leq r1_K \right\}.$$

The constraint can be rewritten as

$$Wu \leq r1_K - b + W\hat{f}(x).$$

Therefore, the inner maximization is the linear program

$$\max_u \left\{ (Q^\top z)^\top u : Wu \leq r\mathbf{1}_K - b + W\hat{f}(x) \right\}.$$

Introducing dual variables $\lambda \in \mathbb{R}_+^K$ associated with the constraints $Wu \leq r\mathbf{1}_K - b + W\hat{f}(x)$, the dual problem is

$$\min_{\lambda \in \mathbb{R}_+^K} \left\{ (r\mathbf{1}_K - b + W\hat{f}(x))^\top \lambda : W^\top \lambda = Q^\top z \right\}.$$

Under strong duality, which holds whenever the inner linear program is feasible and has a finite optimum, the inner maximization is equal to its dual. Hence, the min-max problem is equivalent to

$$\begin{aligned} \min_{W, b, z, \lambda, r, t} \quad & g_0(z) + (r\mathbf{1}_K - b + W\hat{f}(x))^\top \lambda \\ \text{s.t.} \quad & W^\top \lambda = Q^\top z, \quad \lambda \in \mathbb{R}_+^K, \\ & W\epsilon_i + b \leq (r + M(1 - t_i))\mathbf{1}_K, \quad i \in [n], \\ & \mathbf{1}_n^\top t \geq c_n, \\ & z \in \mathcal{Z}, \quad t \in \{0, 1\}^n. \end{aligned}$$

□

A.5 Polyhedral Conformal Set Learning

This section elaborates on the two optimization algorithms described in the main text: the Column-and-Constraint Generation (CCG) method (Alg. 1) for general concave objectives, and the stochastic pinball-subgradient method (Alg. 2) for scalable training.

CCG learning. Alg. 1 solves the problem (1) by iteratively growing a finite scenario set $\mathcal{S}_r \subseteq \mathcal{Y}$. At CCG iteration r , the restricted master problem is:

$$\begin{aligned} \text{MP}_r : \quad & \min_{W, b, z, r, t, \eta} \quad \eta \\ \text{s.t.} \quad & g(z, y^j) \leq \eta, \quad \forall y^j \in \mathcal{S}_r, \\ & W(y^j - \hat{f}(x)) + b \leq r\mathbf{1}_K, \quad \forall y^j \in \mathcal{S}_r, \\ & W\epsilon_i + b \leq (r + M(1 - t_i))\mathbf{1}_K, \quad i \in [n], \\ & \mathbf{1}_n^\top t \geq c_n, \quad z \in \mathcal{Z}, \quad t \in \{0, 1\}^n. \end{aligned} \tag{22}$$

Given a master solution $(\theta^r, z^r, q^r, t^r, \eta^r)$ with $\theta^r = (W^r, b^r)$ and conformal radius q^r , the subproblem finds the worst-case outcome in the current conformal set $\mathcal{C}_{\theta^r}(x, q^r)$:

$$\text{SP}_r : \quad \zeta^r = \max_{y \in \mathcal{Y}} g(z^r, y) \quad \text{s.t.} \quad W^r(y - \hat{f}(x)) + b^r \leq q^r \mathbf{1}_K. \tag{23}$$

When $g(z, \cdot)$ is convex, the maximizer over the polyhedral set can be chosen at an extreme point. CCG terminates when $\zeta^r - \eta^r \leq \varepsilon$, certifying ε -optimality for (12). Because the master problem contains n binary variables $\{t_i\}$, its complexity can grow with n , motivating the pinball surrogate below.

Algorithm 1 CCG for the general concave objective (12)

Require: $\mathcal{D}_n, \varepsilon > 0, M$

- 1: Initialize scenario set $\mathcal{S}_0 \leftarrow \emptyset$, iteration counter $r \leftarrow 0$
 - 2: **repeat**
 - 3: Solve the restricted master problem MP_r in (22)
 - 4: Let $(\theta^r, z^r, q^r, t^r, \eta^r)$ be an optimal master solution
 - 5: Solve the separation problem SP_r in (23)
 - 6: Let ζ^r be its optimal value and y^{r+1} an optimizer
 - 7: **if** $\zeta^r - \eta^r \leq \varepsilon$ **then**
 - 8: **return** $(\theta^r, z^r, q^r, t^r)$
 - 9: **else**
 - 10: Add adversarial scenario: $\mathcal{S}_{r+1} \leftarrow \mathcal{S}_r \cup \{y^{r+1}\}$
 - 11: Set $r \leftarrow r + 1$
 - 12: **end if**
 - 13: **until** termination
-

Algorithm 2 Stochastic pinball-subgradient method for (24)

Require: Learning data $\mathcal{D}_n$, stepsizes $\{\eta_\ell\}$, minibatch size B , calibration penalty γ , $\tau_n = c_n/n$

- 1: Initialize $(\theta^0, r^0, z^0) \in \Theta \times \mathbb{R} \times \mathcal{Z}$, with $\theta^0 = (W^0, b^0)$;
 - 2: **for** $\ell = 0, 1, 2, \dots$ **do**
 - 3: Sample minibatch $\mathcal{B}_\ell \subseteq [n]$, $|\mathcal{B}_\ell| = B$;
 - 4: $y^\ell \in \arg \max_{y \in \mathcal{C}_{\theta^\ell}(x, r^\ell)} g(z^\ell, y)$; $\hat{L}_{\mathcal{B}_\ell}(\theta, r) = \frac{1}{B} \sum_{i \in \mathcal{B}_\ell} \rho_{\tau_n}(S_i(\theta) - r)$;
 - 5: Compute a stochastic subgradient $\hat{G}^\ell$ of $g(z, y^\ell) + \gamma \hat{L}_{\mathcal{B}_\ell}(\theta, r)$ at $(\theta^\ell, r^\ell, z^\ell)$ using (25);
 - 6: $(\theta^{\ell+1}, r^{\ell+1}, z^{\ell+1}) \leftarrow \text{Proj}_{\Theta \times \mathbb{R} \times \mathcal{Z}}((\theta^\ell, r^\ell, z^\ell) - \eta_\ell \hat{G}^\ell)$;
 - 7: **end for**
 - 8: **return** $\hat{\theta} = (\hat{W}, \hat{b})$.
-

Stochastic learning with pinball loss. The exact formulation in (12) enforces empirical coverage through binary variables that select which calibration residuals are contained in the candidate set. To obtain a scalable continuous approximation, we replace this discrete quantile-selection mechanism with a pinball calibration penalty.

For a fixed set parameter θ , write

$$S_i(\theta) := s(X_i, Y_i; \theta), \quad i \in [n],$$

and let $\tau_n := c_n/n$. The empirical conformal radius is the smallest value of r such that at least c_n nonconformity scores are no larger than r . Equivalently, it is the smallest empirical τ_n -quantile of the scores $\{S_i(\theta)\}_{i=1}^n$. This quantile admits the standard pinball-loss characterization. Define

$$\rho_\tau(u) := \tau u_+ + (1 - \tau)(-u)_+, \quad u_+ := \max\{u, 0\},$$

and

$$L_n(\theta, r) := \frac{1}{n} \sum_{i=1}^n \rho_{\tau_n}(S_i(\theta) - r).$$

For fixed θ , the subdifferential with respect to r is

$$\partial_r L_n(\theta, r) = \frac{1}{n} [\#\{i : S_i(\theta) < r\} - \tau_n n, \#\{i : S_i(\theta) \leq r\} - \tau_n n].$$

Hence,

$$0 \in \partial_r L_n(\theta, r) \iff \#\{i : S_i(\theta) < r\} \leq \tau_n n \leq \#\{i : S_i(\theta) \leq r\}.$$

Thus, any minimizer of $L_n(\theta, r)$ is an empirical τ_n -quantile of the nonconformity scores, and the smallest minimizer recovers the conformal radius used in Lemma 2. The pinball loss therefore provides a continuous relaxation of the binary quantile constraint in (12).

We solve the following surrogate problem:

$$\min_{\theta \in \Theta, r \in \mathbb{R}, z \in \mathcal{Z}} \left\{ \max_{y \in \mathcal{C}_\theta(x, r)} g(z, y) + \gamma L_n(\theta, r) \right\}, \quad (24)$$

where $\gamma > 0$ controls the strength of the calibration penalty. The first term learns a polyhedral geometry that is favorable for the downstream robust decision, while the second term keeps the radius aligned with the target empirical conformal quantile. Compared with the exact mixed-integer formulation, this surrogate removes the binary variables and can be optimized with stochastic first-order methods.

For use in Alg. 2, the subgradient of the pinball term with respect to its argument is

$$\xi_i \in \partial \rho_{\tau_n}(S_i(\theta) - r) = \begin{cases} \tau_n, & S_i(\theta) > r, \\ [-(1 - \tau_n), \tau_n], & S_i(\theta) = r, \\ -(1 - \tau_n), & S_i(\theta) < r. \end{cases} \quad (25)$$

Consequently, scores above the current radius push r upward, while scores below the current radius push it downward. When differentiating with respect to θ , the same coefficients ξ_i weight the subgradients of the scores $S_i(\theta)$, allowing the set geometry and radius to be updated jointly.

B Proofs in Section 4

B.1 Proof for Lemma 3

Proof. Since the nonconformity score $s(\cdot, \cdot; \theta)$ is deterministic for fixed θ , and we assume that the augmented sample $\{(X_i, Y_i)\}_{i=1}^{n+1}$ is exchangeable, the induced score sequence $\{s(X_i, Y_i; \theta)\}_{i=1}^{n+1}$ is also exchangeable. Therefore, the rank of $s(X_i, Y_i; \theta)$ among the $n + 1$ scores is uniformly distributed over $\{1, \dots, n + 1\}$. By the definition of the conformal radius $R(\mathcal{D}_n; \theta)$ in (6), there is

$$\mathbb{P}\{S_{n+1} \leq R(\mathcal{D}_n; \theta)\} \geq 1 - \alpha.$$

Using the definition of $\mathcal{C}(x; \mathcal{D}_n, \theta)$ completes the proof:

$$\mathbb{P}\{Y_{n+1} \in \mathcal{C}(X_{n+1}; \mathcal{D}_n, \theta)\} \geq 1 - \alpha.$$

□

B.2 Proof for Lemma 4

Proof. Since $\hat{\theta}(\mathcal{D}_n)$ is learned from $\mathcal{D}_n$, conditioning on $\hat{\theta}(\mathcal{D}_n)$ reveals information about the calibration samples $(X_1, Y_1), \dots, (X_n, Y_n)$, but not about the test point (X_{n+1}, Y_{n+1}) . Consequently, the conditional score sequence

$$s(X_1, Y_1; \hat{\theta}), \dots, s(X_n, Y_n; \hat{\theta}), s(X_{n+1}, Y_{n+1}; \hat{\theta}) \mid \hat{\theta} = \hat{\theta}(\mathcal{D}_n)$$

is generally no longer exchangeable. The loss of exchangeability arises from the asymmetric role of the test point relative to the calibration samples, which is precisely the type of nonexchangeability studied in Barber et al. [2023].

Following the argument of Theorem 2 in Barber et al. [2023], the coverage probability can be lower bounded by the nominal conformal level minus a correction term that measures the deviation from exchangeability. In particular, for the swapped sequence $\mathcal{D}_{n+1}^{(i)}$ defined in the main text, we have

$$\mathbb{P} \left\{ Y_{n+1} \in \mathcal{C}(X_{n+1}, \mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)) \mid \mathcal{D}_n \right\} \geq 1 - \alpha - d_{\text{TV}} \left(\mathcal{D}_{n+1}, \mathcal{D}_{n+1}^{(i)} \mid \mathcal{D}_n \right).$$

Here, the total variation term quantifies the conditional discrepancy between the original augmented sample and the sequence obtained by swapping the test point with a calibration point. When the score sequence is exchangeable, this distance is zero and the usual conformal guarantee is recovered.

Taking expectations with respect to $\mathcal{D}_n$ on both sides yields

$$\mathbb{P} \left\{ Y_{n+1} \in \mathcal{C}(X_{n+1}, \mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)) \right\} \geq 1 - \alpha - \mathbb{E} \left[d_{\text{TV}} \left(\mathcal{D}_{n+1}, \mathcal{D}_{n+1}^{(i)} \mid \mathcal{D}_n \right) \right].$$

This completes the proof. □

B.3 Proof for Theorem 1

Proof. Conditional on the learning data $\mathcal{D}_n$, the learned parameter $\hat{\theta}(\mathcal{D}_n)$ is fixed. By assumption, the re-calibration dataset $\mathcal{D}_m = \{(X_j, Y_j)\}_{j=1}^m$ and the test point (X_{m+1}, Y_{m+1}) are exchangeable with $\mathcal{D}_n$. Therefore, conditional on $\mathcal{D}_n$, the augmented score sequence

$$s(X_1, Y_1; \theta), \dots, s(X_m, Y_m; \theta), s(X_{m+1}, Y_{m+1}; \theta) \mid \theta = \hat{\theta}(\mathcal{D}_n)$$

is exchangeable. Using Lemma 3, there is:

$$\mathbb{P} \left\{ Y_{m+1} \in \mathcal{C}(X_{m+1}; \mathcal{D}_m; \hat{\theta}(\mathcal{D}_n)) \mid \hat{\theta}(\mathcal{D}_n) \right\} \geq 1 - \alpha.$$

Taking expectation with respect to $\mathcal{D}_n$ yields

$$\mathbb{P} \left\{ Y_{m+1} \in \mathcal{C}(X_{m+1}; \mathcal{D}_m, \hat{\theta}(\mathcal{D}_n)) \right\} \geq 1 - \alpha.$$

□

B.4 Proof for Theorem 2

We first introduce the set of regularity conditions that enable finite-sample analysis of the learned parameter and its induced decision. The regularity assumptions are standard in empirical process analysis and primarily serve to control the estimation of the conformal radius and the stability of the induced robust objective.

Assumption 2 (Regularity Conditions). *Assume all data are independent, and:*

- (a) *The function class $\{(x, y) \mapsto \mathbb{1}\{s(x, y; \theta) \leq r\}, \theta \in \Theta, r \in \mathbb{R}\}$ has VC dimension V .*
- (b) *There exists $\kappa > 0$ such that for all $\theta \in \Theta$, $f_\theta(t) \geq \kappa$ for t in a neighborhood of $R^*(\theta)$, where f_θ is the density of the nonconformity scores $s(X, Y, \theta)$.*
- (c) *There exists $L_r \geq 0$ such that for all $\theta \in \Theta$ and $x \in \mathcal{X}$, there is:*

$$|\mathcal{L}(\theta, r) - \mathcal{L}(\theta, r')| \leq L_r \cdot |r - r'|$$

- (d) *Problem (8)–(9) is solved to optimization tolerance ϵ_{opt} .*

The condition (a) controls the complexity of the score class and ensures uniform convergence. The condition (b) guarantees well-behaved quantile estimation near the target radius. The condition (c) ensures stability of the downstream objective with respect to the radius. Finally, the condition (d) accounts for optimization error.

Proof of Theorem 2. We begin by defining the calibration error and the parameter error in (26) and (27), respectively:

$$\Delta(\mathcal{D}_m) = \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R(\mathcal{D}_m; \hat{\theta}(\mathcal{D}_n)); x\right) - \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R^*(\hat{\theta}(\mathcal{D}_n)); x\right), \quad (26)$$

$$\Delta(\mathcal{D}_n) = \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R^*(\hat{\theta}(\mathcal{D}_n)); x\right) - \mathcal{L}\left(\theta^*, R^*(\theta^*); x\right). \quad (27)$$

Then, the suboptimality gap is upper bound by the sum of two errors:

$$\Delta = \Delta(\mathcal{D}_m) + \Delta(\mathcal{D}_n) \leq |\Delta(\mathcal{D}_m)| + \Delta(\mathcal{D}_n).$$

We proceed with analyzing the bound by expanding the two terms respectively.

Calibration error $\Delta(\mathcal{D}_m)$ For any $\theta \in \Theta$, note that the data-driven conformal radius $R(\mathcal{D}_m; \theta)$ is the empirical counterpart of the population quantile $R^*(\theta)$. Denote the cumulative density function (CDF) that they each represent by $\hat{F}_m : \mathbb{R} \rightarrow [0, 1]$ and $F^* : \mathbb{R} \rightarrow [0, 1]$, respectively. Using the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [Dvoretzky et al., 1956], there is:

$$\mathbb{P}\left\{\|\hat{F}_m - F\|_\infty \geq \epsilon\right\} \leq 2e^{-2m\epsilon^2} \Rightarrow \mathbb{P}\left\{\|\hat{F}_m - F\|_\infty \leq \sqrt{\frac{\log(2/\delta)}{2m}}\right\} \geq 1 - \delta.$$

Using (b) in Assumption 2, we know that F has a derivative greater than κ . Since quantile is the inverse function of CDFs, there is a probability greater than $1 - \delta$:

$$|R(\mathcal{D}_m; \theta) - R^*(\theta)| \leq \frac{1}{\kappa} \|\hat{F}_m - F\|_\infty \leq \frac{1}{\kappa} \sqrt{\frac{\log(2/\delta)}{2m}}.$$

By using the Lipschitz condition (c) in Assumption 2 and the arbitrariness of θ , there is:

$$|\Delta(\mathcal{D}_m)| \leq L_r \cdot |R(\mathcal{D}_m; \hat{\theta}(\mathcal{D}_n)) - R^*(\hat{\theta}(\mathcal{D}_n))| \leq \frac{L_r}{\kappa} \sqrt{\frac{\log(2/\delta)}{2m}}.$$

Parameter error $\Delta(\mathcal{D}_n)$ The parameter error $\Delta(\mathcal{D}_n)$ can be further decomposed into three error terms:

$$\Delta(\mathcal{D}_n) = \Delta_1 + \Delta_2 + \Delta_3 \leq |\Delta_1| + \Delta_2 + |\Delta_3|,$$

where

$$\begin{aligned}\Delta_1 &= \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R^*(\hat{\theta}(\mathcal{D}_n)); x\right) - \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R(\mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)); x\right), \\ \Delta_2 &= \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R(\mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)); x\right) - \mathcal{L}\left(\theta^*, R(\mathcal{D}_n; \theta^*); x\right), \\ \Delta_3 &= \mathcal{L}\left(\theta^*, R(\mathcal{D}_n; \theta^*); x\right) - \mathcal{L}\left(\theta^*, R^*(\theta^*); x\right).\end{aligned}$$

The second error term $|\Delta_2|$ can be bounded using condition (d) of Assumption 2:

$$\Delta_2 \leq \mathcal{L}\left(\hat{\theta}(\mathcal{D}_n), R(\mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)); x\right) - \inf_{\theta \in \Theta} \mathcal{L}(\theta, R(\mathcal{D}_n; \theta); x) \leq \epsilon_{\text{opt}}.$$

Using condition (c) of Assumption 2, the first and third error terms share a same upper bound:

$$\begin{aligned}|\Delta_1| &\leq L_r \cdot \left| R^*(\hat{\theta}(\mathcal{D}_n)) - R(\mathcal{D}_n; \hat{\theta}(\mathcal{D}_n)) \right| \leq L_r \cdot \sup_{\theta \in \Theta} |R^*(\theta) - R(\mathcal{D}_n; \theta)| \\ |\Delta_3| &\leq L_r \cdot |R^*(\theta^*) - R(\mathcal{D}_n; \theta^*)| \leq L_r \cdot \sup_{\theta \in \Theta} |R^*(\theta) - R(\mathcal{D}_n; \theta)|.\end{aligned}$$

Since for any θ , the conformal radius $R(\mathcal{D}_n; \theta)$ is an empirical approximation to the population quantile $R^*(\theta)$ using n samples. We redefine the cumulative density function (CDF) that they each represent by $\hat{F}_n(\cdot; \theta) : \mathbb{R} \rightarrow [0, 1]$ and $F^*(\cdot; \theta) : \mathbb{R} \rightarrow [0, 1]$, respectively. Combined with the condition (a) of Assumption 2, using the uniform convergence of VC classes [Vapnik and Chervonenkis, 2015], there is:

$$\mathbb{P}\left\{\sup_{\theta \in \Theta, r \in \mathbb{R}} \left| \hat{F}_n(r; \theta) - F^*(r; \theta) \right| \leq \epsilon\right\} \geq 1 - 8 \left(\frac{en}{V}\right)^V \exp\left(-\frac{n\epsilon^2}{32}\right).$$

By reorganizing the terms, we conclude that with probability greater than $1 - \delta$:

$$\sup_{\theta \in \Theta} \left\| \hat{F}_n(\cdot; \theta) - F^*(\cdot; \theta) \right\|_{\infty} \leq \sqrt{\frac{32 [V \log(en/V) + \log(8/\delta)]}{n}}.$$

Again, using (b) in Assumption 2, we obtain that with probability $1 - \delta$:

$$\sup_{\theta \in \Theta} \left| \hat{R}(\mathcal{D}_n; \theta) - R^*(\theta) \right| \leq \frac{1}{\kappa} \sqrt{\frac{32 [V \log(en/V) + \log(8/\delta)]}{n}}.$$

Therefore, combining the three error terms, there is with probability greater than $1 - \delta$:

$$\Delta(\mathcal{D}_n) \leq \frac{2L_r}{\kappa} \sqrt{\frac{32 [V \log(en/V) + \log(8/\delta)]}{n}} + \epsilon_{\text{opt}}.$$

Combining $\Delta(\mathcal{D}_m)$ and $\Delta(\mathcal{D}_n)$ Combining the three error terms and using a union bound adjustment by setting δ to $\delta/2$ in the previous terms, there is, with probability greater than $1 - \delta$:

$$\Delta \leq \frac{L_r}{\kappa} \sqrt{\frac{\log(4/\delta)}{2m}} + \frac{2L_r}{\kappa} \sqrt{\frac{32[V \log(en/V) + \log(16/\delta)]}{n}} + \epsilon_{\text{opt}}.$$

Using the big- $\mathcal{O}_{\mathbb{P}}$ notation [Van der Vaart, 2000], we conclude that:

$$\Delta \leq \mathcal{O}_{\mathbb{P}} \left(\sqrt{\frac{1}{m}} + \sqrt{\frac{\log n}{n}} \right).$$

□

C Additional Experimental Details and Results

C.1 Experiment details

Data splits and evaluation protocol. All experiments use a fixed four-way split: a training set of 1200 samples for fitting the predictor $\hat{f}$, a learning set $\mathcal{D}_n$ of 300 samples for learning the set geometry θ , a calibration set $\mathcal{D}_m$ of 300 samples for the final conformal radius, and a held-out test set of 1000 samples used only for evaluation. All results are averaged over 5 random seeds, each inducing an independent draw of the full dataset. Baselines that do not use a learning split (Bonferroni, conformal box/ball, PCP) are given the combined $\mathcal{D}_n \cup \mathcal{D}_m$ pool of 600 samples for calibration, ensuring a fair comparison. We set the nominal miscoverage level $\alpha = 0.10$ (i.e., target coverage $1 - \alpha = 0.90$) and $K = 4$ halfplanes in all polyhedral sets.

Synthetic data generation and predictor. For the synthetic tasks, covariates $X \in \mathbb{R}^4$ are drawn from $\mathcal{N}(0, I_4)$, and outcomes $Y \in \mathbb{R}^2$ follow a heteroscedastic nonlinear model:

$$Y = \mu(X) + \varepsilon(X),$$

where $\mu_1(X) = 0.75X_1 + 0.35\sin(X_2) - 0.20X_3$, $\mu_2(X) = -0.45X_1 + 0.55X_2^2 + 0.30X_4$, and $\varepsilon(X)$ is drawn from a zero-mean Gaussian with a covariate-dependent scale factor $0.80 + 0.45\sigma(X_1)$ (where σ is the sigmoid function) applied to a fixed anisotropic root-covariance matrix. An additional 6% heavy-tail contamination is added to each draw. This design produces residuals that are both anisotropic and heteroscedastic, so that the set shape has a meaningful effect on coverage and downstream cost.

We fit the predictor $\hat{f}$ by ridge regression on a nonlinear feature map $\phi(x) = [1, x^\top, x_1^2, x_2^2, \sin(x_1), \sin(x_2)]^\top \in \mathbb{R}^8$, with ridge penalty $\lambda = 10^{-3}$. Residuals $\epsilon_i = Y_i - \hat{f}(X_i)$ on the learning and calibration splits serve as the input to all conformal methods.

Synthetic objectives. We evaluate three two-dimensional robust decision tasks. The decision variable $z \in \Delta_2$ lives on the probability simplex, i.e. $z_1 + z_2 = 1$, $z_j \geq 0$.

Linear objective.

$$g(z, u) = (c + u)^\top z, \quad c = (-0.08, 0).$$

The constant $c_1 = -0.08$ gives the first action a slight structural advantage; the sign of the realized outcome determines whether this advantage holds, making set orientation consequential.

Quadratic-regularized objective.

$$g(z, u) = (c + u)^\top z + \beta(z_1 - z_2)^2, \quad c = (-0.08, 0), \quad \beta = 0.08.$$

The quadratic term penalizes unequal allocation, introducing a smoothly curved Pareto frontier between the two actions.

Newsvendor objective.

$$g(z, u) = \sum_{j=1}^2 [c_o \max(z_j - u_j, 0) + c_u \max(u_j - z_j, 0)], \quad c_o = 0.3, \quad c_u = 0.7.$$

Here $z \in \Delta_2$ is the order quantity and u is the realized demand. Overage ($c_o = 0.3$) and underage ($c_u = 0.7$) costs are asymmetric. The loss is piecewise linear and nonsmooth; its worst-case value depends on which tail of the residual distribution the uncertainty set exposes, making the learned set shape particularly consequential.

Energy Efficiency dataset. We use the UCI Energy Efficiency dataset [Tsanas and Xifara, 2012], also used by Wang et al. [2023] for conformal prediction evaluation. Unlike prior evaluations of this dataset that report only coverage and set volume, we measure downstream worst-case cost and regret. The covariates $X \in \mathbb{R}^8$ encode eight building-design attributes (relative compactness, surface area, wall area, roof area, height, orientation, glazing area, glazing-area distribution), and $Y = (Y_1, Y_2)$ are heating and cooling loads (kWh/m²). We pair the dataset with the linear objective $g(z, u) = (c + u)^\top z$, $c = (-0.08, 0)$, where $z \in \Delta_2$ allocates a unit budget between heating (z_1) and cooling (z_2) control. We fit $\hat{f}$ by the same ridge feature regression as in the synthetic experiments.

C.2 Baseline Implementation Details

We compare against decision-agnostic conformal baselines, including Bonferroni boxes Dunn [1961], conformal boxes Vovk et al. [2005], and conformal balls Wang et al. [2023]. We also include decision-aware baselines based on F-CROMS Bao et al. [2025] and PICNN scores Yeh et al. [2024]. For ablations, “Polyhedron” calibrates a polyhedral set without using the downstream loss, “Polyhedron (min size)” learns the polyhedral set by minimizing volume, and “Ours without re-calibration” learns the decision-aware set using the combined split $\mathcal{D}_n \cup \mathcal{D}_m$ without an independent post-selection calibration step. All methods are evaluated over multiple random seeds using the same train–learn–calibrate–test splits whenever applicable.

Ours. Our method is implemented in two forms. For structured polyhedral templates, we use a CCG-based grid-search implementation. For more scalable learning, we use the stochastic pinball-loss surrogate described in Section 3.2.

F-CROMS. F-CROMS [Bao et al., 2025] selects a conformal prediction set from a finite library of candidate sets, each induced by a different nonconformity score. In the original paper the library consists of model-induced scores; here we instantiate the library with randomly generated halfspace cuts to give the method a fair chance to discover polyhedral geometries similar to ours. $s_k(x, y) = w_k^\top(y - \hat{f}(x))$. For each k we calibrate the threshold on the calibration split via split conformal prediction, obtaining a halfspace-shaped prediction set. F-CROMS then picks the candidate k^* whose set minimizes the empirical worst-case robust objective on the calibration data. We evaluate two library sizes: F-CROMS(10) with 10 candidates and F-CROMS(20) with 20. This implementation is faithful to the original method’s spirit: it can only select among a discrete pool of pre-specified directions and cannot optimize the halfspace orientation continuously. Our method, by contrast, jointly optimizes the halfspace parameters W and b over a continuous search space, which strictly subsumes any finite candidate library and yields a smaller decision risk whenever the optimal geometry is not representable by a randomly drawn cut.

Yeh et al. Yeh et al. [2024] trains the uncertainty set end-to-end by parameterizing the nonconformity score via a partially input-convex neural network (PICNN), which guarantees that the sublevel set $\{y : s(x, y; \theta) \leq \tau\}$ is convex in y for every x . Gradients with respect to θ flow through the adversarial argmax via KKT conditions and implicit differentiation, enabling joint learning of set shape and size. We implement this baseline using the authors’ PICNN score class with a two-hidden-layer architecture (width 64, ReLU activations) and optimize with Adam (step size 10^{-2} , 200 epochs). The conformal radius is then calibrated on the same held-out split as ours so that the final coverage guarantee is comparable. This design is sensible when $g(z, \cdot)$ is differentiable and the gradient signal can propagate through the PICNN; it produces general convex sets and is thus potentially more expressive than our polyhedral parameterization.

C.3 Additional Results

Tables 1 and 2 provide additional results. Across the three synthetic objectives, the proposed re-calibrated method maintains empirical coverage close to the nominal 90% level while achieving the best or near-best worst-case cost. In the linear and quadratic-regularized objectives, our method substantially improves worst-case cost relative to standard conformal boxes, conformal balls, PCP, and fixed polyhedral baselines. For the newsvendor task, conformal ℓ_2 attains the lowest worst-case cost, but our method remains competitive while using a decision-aware learned geometry. These results further show that smaller volume alone does not necessarily imply better downstream performance: PCP and the minimum-size polyhedron often produce compact sets but can incur larger worst-case costs.

The real-data energy-efficiency experiment in Table 2 shows a similar pattern. Our method achieves the lowest worst-case cost among all methods, while maintaining coverage near the target level within sampling variability. Compared with Ours (w/o recal), the re-calibrated version has nearly identical regret and runtime, but provides the statistically valid two-stage construction used in the main method. Overall, these results support the conclusion that learning the geometry of the conformal set in a decision-aware manner improves robust downstream performance without sacrificing empirical coverage.

Table 1: Comparison across synthetic objectives. Mean $\pm$ std over five seeds; $n = m = 300$, $K = 4$, $1 - \alpha = 0.90$. WC cost and regret are in units of $\times 10^{-1}$; time is total wall-clock seconds per run.

Task	Method	Coverage	Volume	WC cost	Regret	Time (s)
Linear objective	Bonferroni box	0.913 ± 0.021	0.84 ± 0.15	1.35 ± 0.26	0.26 ± 0.16	< 0.01
	Conformal (ℓ_∞)	0.901 ± 0.021	1.10 ± 0.24	1.64 ± 0.51	0.11 ± 0.02	< 0.01
	Conformal (ℓ_2)	0.896 ± 0.025	1.03 ± 0.12	1.97 ± 0.21	0.30 ± 0.06	< 0.01
	PCP	0.915 ± 0.021	0.68 ± 0.07	1.83 ± 0.40	0.33 ± 0.22	< 0.01
	F-CROMS (10)	0.897 ± 0.023	0.86 ± 0.10	1.35 ± 0.27	0.27 ± 0.15	0.13
	F-CROMS (20)	0.899 ± 0.024	0.85 ± 0.08	1.31 ± 0.24	0.30 ± 0.16	0.25
	Yeh et al.	0.899 ± 0.023	1.10 ± 0.08	2.04 ± 0.45	0.24 ± 0.06	4.6
	Polyhedron	0.906 ± 0.022	0.69 ± 0.13	2.23 ± 0.88	0.35 ± 0.08	< 0.01
	Polyhedron (min size)	0.909 ± 0.027	0.61 ± 0.09	2.44 ± 0.73	0.29 ± 0.11	< 0.01
	Ours (w/o recal)	0.901 ± 0.019	0.78 ± 0.15	1.15 ± 0.34	0.27 ± 0.17	1.6
	Ours	0.902 ± 0.021	0.78 ± 0.04	<u>1.19 ± 0.44</u>	0.29 ± 0.24	1.6
Quadratic-regularized objective	Bonferroni box	0.913 ± 0.021	0.84 ± 0.15	2.10 ± 0.27	0.24 ± 0.14	< 0.01
	Conformal (ℓ_∞)	0.901 ± 0.021	1.10 ± 0.24	2.39 ± 0.51	0.12 ± 0.03	< 0.01
	Conformal (ℓ_2)	0.896 ± 0.025	1.03 ± 0.12	2.55 ± 0.21	0.29 ± 0.04	< 0.01
	PCP	0.915 ± 0.021	0.68 ± 0.07	2.51 ± 0.42	0.31 ± 0.21	< 0.01
	F-CROMS (10)	0.897 ± 0.023	0.86 ± 0.10	2.06 ± 0.26	0.26 ± 0.14	0.50
	F-CROMS (20)	0.899 ± 0.024	0.85 ± 0.08	2.01 ± 0.24	0.28 ± 0.15	0.97
	Yeh et al.	0.899 ± 0.023	1.10 ± 0.08	2.70 ± 0.44	0.24 ± 0.03	4.5
	Polyhedron	0.906 ± 0.022	0.69 ± 0.13	2.87 ± 0.83	0.33 ± 0.05	< 0.01
	Polyhedron (min size)	0.909 ± 0.027	0.61 ± 0.09	3.09 ± 0.71	0.28 ± 0.08	< 0.01
	Ours (w/o recal)	0.906 ± 0.016	0.81 ± 0.09	<u>1.89 ± 0.35</u>	0.20 ± 0.07	5.9
	Ours	0.901 ± 0.019	0.78 ± 0.04	1.80 ± 0.32	0.20 ± 0.08	5.9
Newsvendor	Bonferroni box	0.911 ± 0.018	0.88 ± 0.22	8.76 ± 0.51	0.21 ± 0.04	< 0.01
	Conformal (ℓ_∞)	0.898 ± 0.014	1.08 ± 0.20	9.07 ± 0.37	0.19 ± 0.03	< 0.01
	Conformal (ℓ_2)	0.895 ± 0.016	1.05 ± 0.07	8.03 ± 0.30	0.15 ± 0.05	< 0.01
	PCP	0.910 ± 0.027	0.69 ± 0.10	10.63 ± 0.82	5.55 ± 0.08	< 0.01
	F-CROMS (10)	0.901 ± 0.019	0.77 ± 0.20	8.30 ± 0.52	0.22 ± 0.08	0.49
	F-CROMS (20)	0.903 ± 0.023	0.81 ± 0.22	8.30 ± 0.51	0.22 ± 0.08	0.96
	Yeh et al.	0.895 ± 0.019	1.10 ± 0.10	8.68 ± 0.28	0.24 ± 0.08	6.0
	Polyhedron	0.898 ± 0.025	0.73 ± 0.18	8.39 ± 0.40	0.24 ± 0.04	< 0.01
	Polyhedron (min size)	0.901 ± 0.029	0.59 ± 0.12	8.47 ± 0.46	0.26 ± 0.09	< 0.01
	Ours (w/o recal)	0.892 ± 0.025	0.70 ± 0.22	<u>8.05 ± 0.50</u>	0.20 ± 0.09	6.5
	Ours	0.898 ± 0.017	0.73 ± 0.19	8.12 ± 0.38	0.21 ± 0.12	5.8

Table 2: Energy Efficiency (real data). Mean $\pm$ std over five seeds; $n = m = 300$, $K = 4$, $1 - \alpha = 0.90$. WC cost and regret are in units of $\times 10^{-1}$; time is total wall-clock seconds per run.

Method	Coverage	Volume	WC cost ($\times 10^{-1}$)	Regret ($\times 10^{-1}$)	Time (s)
Bonferroni box	0.914 ± 0.017	0.72 ± 0.08	2.29 ± 0.97	0.43 ± 0.12	< 0.01
Conformal (ℓ_∞)	0.915 ± 0.023	0.88 ± 0.23	3.53 ± 1.26	0.32 ± 0.03	< 0.01
Conformal (ℓ_2)	0.908 ± 0.033	0.84 ± 0.12	2.94 ± 1.02	0.74 ± 0.05	< 0.01
PCP	0.906 ± 0.027	0.65 ± 0.09	3.41 ± 0.84	0.41 ± 0.10	< 0.01
F-CROMS (10)	0.892 ± 0.033	0.83 ± 0.24	2.01 ± 0.92	0.56 ± 0.18	0.13
F-CROMS (20)	0.893 ± 0.034	0.85 ± 0.24	1.93 ± 0.93	0.54 ± 0.17	0.25
Yeh et al.	0.914 ± 0.022	0.89 ± 0.21	3.24 ± 1.24	0.40 ± 0.06	2.7
Polyhedron	0.900 ± 0.023	0.66 ± 0.11	3.70 ± 1.22	0.84 ± 0.10	< 0.01
Polyhedron (min size)	0.929 ± 0.020	0.65 ± 0.13	3.06 ± 1.34	0.77 ± 0.18	< 0.01
Ours (w/o recal)	0.897 ± 0.036	1.20 ± 0.24	<u>1.80 ± 0.83</u>	0.48 ± 0.15	1.6
Ours	0.896 ± 0.024	1.16 ± 0.17	1.76 ± 0.88	0.48 ± 0.15	1.6